Three current structural tables over $\mathbb{F}_4$

This finite application certifies every displayed subgroup description in the current Connor–Leemans tables for $\mathrm{PSL}_3(4)$, $\mathrm{PGL}_3(4)$ and $\mathrm{P\Sigma L}_3(4)$. The complete subgroup inventories, exact source-row correspondences and corrected maximal-incidence weights are acquired from the accepted numerical components. This paper adds the structural descriptions; it repeats no ambient subgroup census.

Exact assertion. Relative to the acquired named models and earlier structural certificates, each of the 347 source rows admits its displayed decomposition. The literal certificates identify the actual embedded subgroup in that row, not merely another group of the same order. Extension descriptions need not specify a unique isomorphism type: the assertion is the existence of the displayed normal subgroup, quotient, complement or direct factors, with nonsplitting where a dot is printed.

Current companion table	Rows	Actual degree	Native pages
$\mathrm{PSL}_3(4)$	95	21	4
$\mathrm{PGL}_3(4)$	100	21	4
$\mathrm{P\Sigma L}_3(4)$	152	44	7
Total	347		15

All fifteen native pages were directly read, including the last field-extension page containing the continued-table heading and attribution. Powers, dots, colons, products and association are retained in a separate syntax tree for each row. For example, field-extension row 22 is $2^{1+4} \cdot 2^2$; rows 12–13 end with a dot, $((2^4 : 5) : 2) \cdot 2$. The final product in projective-linear row 34 is $(7 : 3) \times 3$. Flattened extracted digits are interpreted only after this native check.

Named and embedded inputs. The named $L_3(4)$ atom is the exact accepted 21-point projective construction already acquired in the M22 structural component from the L100 construction packet. Its identity is not inferred from the order $20,160$. The first two numerical files contain only literal permutation groups; every proper subgroup retains its original 21-point embedding. The field extension uses the accepted 44-point literal restriction state and its exact field-automorphism identification. No display-name constructor replaces an embedded group. The two degrees are deliberately kept separate: every transport states its actual target degree and its generator images.

There are 140 acquired registry entries, including cyclic, elementary abelian, dihedral, quaternion, extraspecial, Mathieu and projective models where they actually occur in a source factor. The present witness discovery adds only eleven models: ten semidirect products and one direct product. All nonsplit descriptions in this roster are transported from previously accepted models, including the complete quotient-lift obstructions.

Complete finite verification. We recall the acquired labelled-generator transport criterion. List every element of a finite source model H and its products by each ordered generator. A bijection f onto the actual target subgroup, fixing identity and satisfying $f(hs) = f(h)f(s)$ on every such edge, is a homomorphism: induction on generator words gives the equation for multiplication by each word. Hence it is an isomorphism. The verifier checks the entire bijection, all edges and actual generator membership. Candidate isomorphism searches merely propose these maps; their answers are never accepted without the literal checks.

For a new split model the certificate gives actual subgroups $N, K \leq H$. The finite reader checks their already certified names, normality of N on the actual generators of H, $N \cap K = 1$ and $|N||K| = |H|$. Thus $H = NK$ and the asserted split extension follows. It also checks a noncommuting generator pair for a printed nontrivial semidirect action. For the one direct product all factor generators commute instead; generation and trivial intersection give $H \cong N \times K$. The model-reference graph is checked in dependency order, so none of the new names is its own premise.

Every row is joined to its retained numerical representative before a certificate is considered. The reader checks the exact source row/order, the native syntax tree, the complete source-to-computation bijection, the literal original generators and the actual action degree. It then enumerates that finite target subgroup by permutation multiplication and checks the complete transport. In particular the top-row split assertions are certified just like the proper rows. All 347 occurrences are required; there is no implicit credit for a repeated description.

Evidence and limits. The bounded discovery returned all 347 witnesses in 6.27 seconds without any missing row. Only the resulting literal arrays are used by the independent Python reader. Ordinary and optimized executions must produce identical reports. Damage controls target the source roster, dots, powers, embedded generators, degrees, named-display substitutions and complete-state marker. A separate $Q_8 \to C_2^2$ example checks the acquired complete-coset nonsplitting guard and rejects a duplicated quotient coset.

The source attribution is Connor and Leemans, the current companion files associated with *An atlas of subgroup lattices of finite almost simple groups*; these files print November 26, 2012. Their exact current bytes are bound. No identity with a particular historical 2013 or 2015 file edition is asserted. Previously accepted numerical and named-constructor interfaces remain explicit inputs. This application makes no novelty, full-atlas, whole-original, conjecture, formalization or measurement claim.